

SARVARBEK ERNIYAZOV

Ph.D. Candidate, Computer Engineering

Time-Series Forecasting, Anomaly Detection and Reliable Decisions under Distribution Shift

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[Google Scholar](#) | [GitHub](#) | [Research Portfolio](#) | [Hugging Face](#) | [LinkedIn](#)

Academic Profile

I work on time-series forecasting and anomaly detection for energy and cyber-physical systems, and on when such models can be trusted under distribution shift. My peer-reviewed work (three lead-author journal articles, 2024–2026) covers graph-based and multimodal solar forecasting and industrial-control anomaly detection. In public, documented studies I have tested conformal intervals, abstention and drift detectors under campaign and site shift, and recorded where they fail. My next aim is risk-controlled act-or-abstain decisions that stay valid when sites, sensors or seasons change.

Research Focus

Conformal prediction and risk control under distribution shift • Selective prediction and abstention • Probabilistic, multi-horizon time-series forecasting for energy • Anomaly and drift detection with false-alarm control • Learning across heterogeneous sites and clients

Education and Availability

Ph.D. in Computer Engineering, Chonnam National University, Yeosu, South Korea 2020 – Feb. 2027 (exp.)

Dissertation (working title): *Shift-Resilient Learning for Heterogeneous Energy Systems: From Spatio-Temporal Forecasting to Risk-Calibrated Event Prediction*

Advisor: Prof. Chang Gyoong Lim • Dissertation defense: **December 2026** • Degree conferral: **February 2027**

M.S. in Computer Engineering, Kumoh National Institute of Technology, Gumi, South Korea 2016 – 2018

Information Security Laboratory (QCA Research Group)

B.S. in Computer Engineering (Information Security), Tashkent University of Information 2011 – 2015

Technologies, Uzbekistan

Postdoctoral availability: from March 2027, and compatible with later start dates. My university affiliation and grant responsibilities conclude upon degree conferral in February 2027; the funded project continues to 2028, but my personal participation ends at conferral.

Record at a Glance

- **9 peer-reviewed publications** (7 journal articles, 2 proceedings papers), lead author on all; **3 are machine-learning journal articles** (2024–2026). One further manuscript is under review and listed separately.
- **124 citations** across the full record (Google Scholar, 11 September 2026), including **93** to the 2019 *Microelectronic Engineering* article on low-power circuits.
- **11 public research repositories** and **7 hosted research interfaces**, including tested code (continuous integration in DriftSentinel).
- Korean national R&D in virtual power plants, edge–cloud energy management and industrial monitoring; on the active TIPA project, Lead Researcher for the assigned university AI work package; not the formal grant PI.

Uncertainty, Risk-Control and Shift Studies (public code; not peer-reviewed)

- **CellTriage** — Risk-controlled battery quality decisions. Early-cycle life prediction with conformal intervals and cost-aware accept/reject/defer decisions (Extra Trees: MAPE $9.57\% \pm 3.35$, $\rho = 0.944$; $14.2\times$ cost reduction versus AQL sampling, conditional on the stated cost matrix). Reported conformal coverage falls from about 90% in distribution to 60% under held-out campaign shift, documenting failure of nominal coverage when exchangeability is violated. [Code](#) • [Research console](#)
- **DriftSentinel** — Self-auditing distribution-shift monitoring. Executable comparison of PSI, KS and chi-square tests with BH-FDR correction, CUSUM, MMD and BBSD on a 101,766-encounter clinical model. 20-seed negative controls identify false drift alarms and invalid detector behavior and overturn an initial “8/8 critical” result. 93 automated tests, continuous integration and determinism checks. [Code](#) • [Hosted monitoring console](#)
- **FaultLine** — Sole-author, pre-registered study and the final study of my dissertation: a from-scratch joint telemetry–text sequence model for wind-turbine event risk (four public SCADA sources, one 8 GB GPU). The result is inconclusive. The registered programme closed on 20 September 2026 after six gates; two follow-up gates followed. A second test, registered after an inconclusive initial test, puts the joint model $+0.017$ to $+0.023$ AUPRC above telemetry-only on every seed on the registered label (inconclusive under a label variant), level with an order-blind classifier of the status strings; post-closure, a time-since-last-fault baseline beats it (exploratory). Two site-shift tests were negative; abstention was evaluated, graceful degradation not established. Not an operational system. [Repository](#)

Research Experience

Ph.D. Researcher — AI Laboratory, Chonnam National University

2020 – present

Department of Computer Engineering, Yeosu, South Korea • Advisor: Prof. Chang Gyoong Lim

Research program: spatio-temporal, multimodal, and distributed learning for energy forecasting and cyber-physical anomaly detection.

- Led, as lead author, the research on graph-enhanced temporal and frequency modeling for multi-station solar forecasting, including architecture design, implementation, baseline comparison, ablation analysis, and manuscript preparation.
- Developed dynamic graph and mixture-of-experts methods for integrating heterogeneous solar and meteorological modalities across forecast horizons.
- Investigated federated spatio-temporal mixture-of-experts for heterogeneous building-energy clients; the manuscript is under review and is excluded from the peer-reviewed publication count.
- Developed and evaluated autoencoder-based time-series anomaly detection on the HAI industrial-control-system benchmark.
- Built reproducible research pipelines: data validation, leakage-aware splitting, training, statistical evaluation, ablations, and research-prototype integration.
- Contributed AI methodology and technical deliverables to Korean national R&D programs involving virtual power plants, edge-cloud energy management and power-infrastructure monitoring, and methodology to a submitted proposal on satellite-ground data integration.

Individual contribution statement: For my lead-author doctoral studies, I led method development, software implementation, experimental design, baseline and ablation evaluation, result visualization, and manuscript preparation. Formal grant governance and institutional PI responsibilities remained with the appointed faculty principal investigator.

Graduate Researcher — Information Security Laboratory / QCA Research Group

2016 – 2018

Kumoh National Institute of Technology, Gumi, South Korea

- Designed and simulated low-power quantum-dot cellular automata arithmetic and reversible circuits; reported more than 50% area reduction for the proposed carry-look-ahead architecture relative to the specified comparison design; lead author of the 2019 *Microelectronic Engineering* article.

National R&D Projects

On the funded projects below, grant governance remained with the appointed faculty PI.

- **TIPA RS-2026-25529976 — sLM-LMM Cooperative Virtual Power Plant Platform** (funded, active, 2026–2028; industry-academia commercialization R&D). **Role:** Lead Researcher for the assigned university AI work package; not the formal grant PI. Model and evaluation design for cooperation between on-device small language models and cloud multimodal models; AI-side interfaces for telemetry, forecasts and operator tools; technical validation of assigned machine-learning deliverables. My participation ends at degree conferral in February 2027.
- **TIPA RS-2025-02312851 — Preliminary study for an on-device AI and cloud-integrated VPP platform** (funded, completed, 2025). **Role:** Lead AI Researcher for the machine-learning feasibility work; not the formal grant PI. AI feasibility analysis, model architecture planning and prototype evaluation for the follow-on project.
- **Submitted proposals, not funded awards:** NRF RS-2025-00564485 (satellite-ground data integration for solar variability management and VPP operations; methodology contributor) and INNOPOLIS RS-2026-25550513 (AI-based power-equipment anomaly detection and preventive control; methodology contributor).
- **Earlier funded programmes, as research contributor:** NRF/MSIP NRF-2017R1D1A3B03034346; MOE Regional Innovation Strategy 2021RIS-002; MSS RS-2023-00278623.

Publications

Verified publication records are listed below, with manuscripts under review reported separately and excluded from peer-reviewed publication counts. Citation counts were confirmed on Google Scholar on 11 September 2026. My name is shown in bold. Full record: [Google Scholar](#) and [ORCID](#).

Record summary 7 peer-reviewed journal articles • 2 peer-reviewed proceedings papers • 1 manuscript under review • lead author on every entry

Machine Learning — Peer-Reviewed Journal Articles (2024–2026)

- [J1] **S. Erniyazov**, M. A. Jaleel, C. G. Lim, and S. B. Ha. “A multimodal fusion framework for solar forecasting using dynamic GNNs and mixture of experts.” *KSII Transactions on Internet and Information Systems*, 20(3), 1337–1360, 2026. doi:10.3837/tiis.2026.03.012
Contribution: lead-author multimodal architecture, dynamic graph and expert-routing methodology, evaluation, analysis, and manuscript preparation.
- [J2] **S. Erniyazov** and C. G. Lim. “GNN-enhanced temporal patch segmentation and frequency fusion model for robust solar energy production forecasting.” *Energy Reports*, 13, 4962–4984, 2025. doi:10.1016/j.egy.2025.03.063
Contribution: lead-author method development, implementation, experimental design, baseline and ablation evaluation, and manuscript preparation.

- [J3] **S. Erniyazov**, Y.-M. Kim, M. A. Jaleel, and C. G. Lim. “Comprehensive analysis and improved techniques for anomaly detection in time series data with autoencoder models.” *International Journal on Advanced Science, Engineering and Information Technology*, 14(6), 1861–1867, 2024. doi:10.18517/ijaseit.14.6.20451
Contribution: lead-author model design, HAI benchmark evaluation, analysis, and manuscript preparation.

Manuscript Under Review — Not Included in Publication Count

- [M1] **S. Erniyazov** et al. “Fed-ST-MoE: A federated spatio-temporal GNN with mixture-of-experts for heterogeneous building energy and demand forecasting.” Manuscript under review.
Investigates decentralized forecasting across heterogeneous buildings with spatio-temporal graph learning and client-aware mixture-of-experts. The study addresses statistical heterogeneity and local-data retention; it makes no formal privacy claim. Venue and preprint link will be disclosed as permitted by the review policy.

Earlier Research — Low-Power QCA Circuits (2017–2019)

- [J4] **S. Erniyazov** and J.-C. Jeon. “Carry save adder and carry look ahead adder using inverter chain based coplanar QCA full adder for low energy dissipation.” *Microelectronic Engineering*, 211, 37–43, 2019. doi:10.1016/j.mee.2019.03.015
93 citations (Google Scholar, 11 September 2026). Contribution: circuit design, simulation, comparative evaluation, and manuscript preparation.
- [J5] **S. Erniyazov** and J. C. Jeon. “Coplanar subtractor design in QCA for arithmetic circuit design.” *Advanced Science Letters*, 23(10), 10112–10117, 2017. doi:10.1166/asl.2017.10399
- [J6] **S. Erniyazov** and J. C. Jeon. “Implementation of area efficient Fredkin gate for compact reversible QCA circuit.” *Advanced Science Letters*, 23(10), 10092–10096, 2017. doi:10.1166/asl.2017.10395
- [J7] **S. Erniyazov** and J. C. Jeon. “Reversible circuit design in QCA based on double Feynman gate.” *Advanced Science Letters*, 23(10), 9852–9856, 2017. doi:10.1166/asl.2017.9810
- [C1] **S. Erniyazov** and J.-C. Jeon. “Area efficient magnitude comparator based on QCA.” *Advanced Science and Technology Letters*, 150, 75–79, 2018. doi:10.14257/asl.2018.150.19
- [C2] **S. Erniyazov** and J.-C. Jeon. “Half subtractor in QCA for arithmetic circuits.” In *Proceedings of the International Workshop on Future Technology (FUTECH 2017)*, Korean Institute of Information Technology, pp. 148–149, July 2017. [Proceedings record](#)

Studies in Progress — Not Publications

No peer-reviewed output is claimed for either study.

- **AudioShield** — exploratory cross-corpus anti-spoofing study (WavLM, XLS-R) under codec, resampling, replay and corpus shift; six controlled experiments and a domain-reliance analysis are public. Blind evaluation pending; no final performance claim. Involves external collaborators, who will be named once public attribution is agreed. [Code](#)
- A further empirical measurement study is in preparation for double-blind submission; details are withheld to preserve anonymity.

Other Public Artifacts

Code, model weights or demos; none is peer-reviewed or in operational use. WaveSeg (770-parameter frequency-boundary adapter for SegFormer-B0; reported LGG HD95 29.70 → 22.83 pixels) • FreqFD (cross-generator deepfake detection; unseen-generator AUC 0.588 → 0.686, generalization gap acknowledged) • ReproLab (AI4I-2020 reproducibility audit; F1 ceiling 0.9558 derived three ways) • DALM-DETR • OCRipple • FinRiskGuard • Auto Damage Segmentation • Uzbek Handwriting OCR. Index: sarvarbek-erniyazov.github.io

Methods

Conformal prediction and risk control; uncertainty quantification and calibration; selective prediction; probabilistic and multi-horizon time-series forecasting; anomaly and drift detection; spatio-temporal graph learning; multimodal fusion and mixture-of-experts; federated and personalized learning; frequency-domain learning; computer vision.

Software: Python; PyTorch; scikit-learn; Git with automated testing and continuous integration; Docker; FastAPI; Hugging Face; Streamlit.

Open Science and Academic Service

- Maintainer of public reproducibility artifacts, model cards, and research demonstrations supporting transparent evaluation (leakage audits, negative controls, multi-seed reporting and automated tests across the artifacts listed above; continuous integration in DriftSentinel).
- Laboratory service at the AI Laboratory, Chonnam National University: research-pipeline and evaluation-protocol support for national R&D deliverables.